

OS

Assignment-2

Problem Statement -

Write a C program for sparse matrix realization and operations on it - Transpose, Fast Transpose.

Objective:

1. To study the concept of sparse matrix, how it's stored and displayed.
2. To understand the implementation of sparse matrix operation - Simple Transpose, Fast Transpose.

Implementation:

Platform:

- 64 bit Open source linux or its derivatives
- Open source C programming tool like gcc/Eclipse Editor

Pseudo Code:

1. Input Matrix to Compact (Triplet) form

Begin

Read number of rows (m) and columns (n)

Read matrix $A[m][n]$

count $\leftarrow 0$

For $i \leftarrow 0$ To $m-1$

For $j \leftarrow 0$ To $n-1$

If $A[i][j] \neq 0$ Then

Count $\leftarrow$ Count + 1

Triplet [Count][0] $\leftarrow i$

Triplet [Count][1] $\leftarrow j$

Triplet [Count][2] $\leftarrow A[i][j]$

END IF

END FOR

END FOR

Triplet [0][0] $\leftarrow m$

Triplet [0][1] $\leftarrow n$

Triplet [0][2] $\leftarrow$ Count

Display Triplet Matrix

END

2. ~~Pr.~~ Simple Transpose

BEGIN

Read Triplet Matrix T

Transpose [0][0] $\leftarrow T[0][1]$

Transpose [0][1] $\leftarrow T[1][0]$

Transpose [0][2] $\leftarrow T[2][0]$

$col = 1$

For $col = 0$ TO $T[0][1] - 1$

FOR $i = 1$ TO $T[0][2]$

IF $T[i][1] = col$ THEN

Transpose $[k][0] \leftarrow T[i][1]$

Transpose $[k][1] \leftarrow T[i][0]$

Transpose $[k][2] \leftarrow T[i][2]$

$k \leftarrow k + 1$

END IF

END FOR

END FOR

Display Transpose Matrix

END

3. Fast Transpose

BEGIN

Read Triplet Matrix T

Transpose $[0][0] \leftarrow T[0][1]$

Transpose $[0][1] \leftarrow T[0][0]$

Transpose $[0][2] \leftarrow T[0][2]$

Initialize rowTerm[] to 0

$R \leftarrow 1$

For $col \leftarrow 0$ To $T[0][1] - 1$

FOR $i \leftarrow 1$ TO $T[0][2]$

IF $T[i][1] = col$ THEN

Transpose $[R][0] \leftarrow T[i][1]$

Transpose $[R][1] \leftarrow T[i][0]$

Transpose $[R][2] \leftarrow T[i][2]$

$R \leftarrow R + 1$

END IF

END FOR

END FOR

Display Transpose Matrix

END

3. Fast Transpose

BEGIN

Read Triplet Matrix T

Transpose $[0][0] \leftarrow T[0][1]$

Transpose $[0][1] \leftarrow T[0][0]$

Transpose $[0][2] \leftarrow T[0][2]$

Initialize rowTerm[] to 0

Initialize startPos[] ~~to 0~~ to 0

FOR $i \in 1$ TO $T[0][2]$

 rowTerms $[T[i][1]] \leftarrow$ rowTerms $[T[i][1]]$

END FOR

startPos $[0] \leftarrow 1$

FOR $i \in 1$ TO $T[0][1] - 1$

 startPos $[i] \leftarrow$ startPos $[i-1] +$ rowTerms

END FOR

FOR $i \in 1$ TO $T[0][2]$

 position $\leftarrow$ startPos $[T[i][1]]$

 Transpose[position] $[0] \leftarrow T[i][1]$

 Transpose[position] $[1] \leftarrow T[i][0]$

 Transpose[position] $[2] \leftarrow T[i][2]$

startPos $[T[i][1]] \leftarrow$ startPos $[T[i][1]]$

END FOR

Display Transpose Matrix

END

• Time Complexity:

Simple Transpose = $O(\text{columns} \times \text{non-zero elements})$

Fast Transpose = $O(\text{columns} + \text{non-zero elements})$

• Conclusion:

Thus, implemented sparse matrix operations ~~are~~ assignment. This System is able to perform different operations on sparse matrices such as simple and fast transpose and their time complexities.

FAQs

- What is a sparse matrix? list the applications?
- Represent sparse matrices with suitable data structures? Explain with an example simple and Fast Transpose?

a) A Sparse matrix is a matrix in which most of the elements are zero. Instead of storing every element, only the non-zero elements are stored.

Applications:

- Graph representation
- GIS
- Network analysis

- Image processing
- Machine Learning

b) Sparse matrices can be represented using:

1. Triplet Representation
2. Linked list Representation
3. Array of structures

Simple Transpose

The matrix is transposed by ~~scanning~~ scanning column one by one and copying matching elements into the result matrix. It requires repeated scanning, making it slower.

Fast Transpose -

The number of elements in each column is counted first. Their starting positions are calculated allowing each element to be placed directly in its correct position. This makes the algorithm faster.

Q. Find out the addition of 2 sparse matrices in form and also find Simple and Fast Transpose?

$$M_1 = \begin{array}{ccc} 4 & 5 & 6 \\ 0 & 3 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 45 \\ 2 & 3 & 4 \\ 3 & 2 & 45 \\ 4 & 1 & 2 \end{array}$$

$$M_2 = \begin{array}{ccc} 4 & 5 & 6 \\ 0 & 3 & 7 \\ 0 & 4 & 6 \\ 1 & 4 & 4 \\ 2 & 1 & 8 \\ 3 & 2 & 45 \\ 4 & 4 & 21 \end{array}$$

$$M_1 + M_2 = \begin{array}{ccc} 4 & 5 & 8 \\ 0 & 3 & 12 \\ 0 & 4 & 6 \\ 1 & 4 & 49 \\ 2 & 1 & 8 \\ 2 & 3 & 4 \\ 3 & 2 & 90 \\ 4 & 1 & 2 \\ 4 & 4 & 21 \end{array}$$

Simple Transpose of M_1

$$\begin{array}{ccc} 5 & 4 & 6 \\ 1 & 4 & 2 \\ 2 & 3 & 45 \\ 3 & 0 & 5 \\ 3 & 2 & 4 \\ 4 & 1 & 45 \end{array}$$

Fast Transpose of M_1

$$\begin{array}{ccc} 5 & 4 & 6 \\ 1 & 4 & 2 \\ 2 & 3 & 45 \\ 3 & 0 & 5 \\ 3 & 2 & 4 \\ 4 & 1 & 45 \end{array}$$